



The Family Photo Album Mystery!



Grade 4 Science • Traits, Inheritance & Genetics

Ananya's Discovery

Ananya was flipping through an old family photo album with her grandfather (Dada) and grandmother (Dadi). She noticed something amazing: "Why do I have dimples when I smile, just like Dadi? And why are my brother's eyes dark brown like Papa's?" Let's solve the mystery of **Traits**!



What Are Traits?

Traits are special physical features or qualities that make you unique and show how you are connected to your family.

- Eye color (Brown, Light brown, Green)
- Hair type (Curly, Straight, Wavy)
- Dimples & Free vs. Attached Earlobes



Inherited vs. Learned Traits

Inherited Traits: Features passed down from parents & grandparents through genetics (e.g., Brown eyes, Curly hair).

Learned Traits: Skills you acquire by practicing or learning from relatives (e.g., Playing Tabla, Dancing Bharatanatyam, Cooking Dosa).



Meet Gregor Mendel: The Trait Detective!

Gregor Mendel was a scientist who studied how traits pass from parents to children using pea plants. Just like mango trees in an orchard can produce sweet or sour mangoes, traits come in different versions!



Dominant Traits (Loud & Proud)

These traits show up more often! Like bossy trees in a mango orchard.

Examples:

Brown Eyes

Curly Hair

Dimples

Free Earlobes



Recessive Traits (Quiet & Hidden)

These traits stay hidden unless passed down from both parents!

Examples:

Light/Blue Eyes

Straight Hair

No Dimples

Attached Earlobes



Case Study: The Verma Family

In the Verma family:

- **Dada:** Free earlobes (Dominant)
- **Dadi & Parents:** Attached earlobes (Recessive)
- **Children:** Mix of both types across generations!



Self-Check: Discover Your Traits!

- **Tongue Rolling:** Can you roll your tongue into a 'U' shape?
- **Earlobes:** Are they free-hanging or attached to the side of your head?
- **Dimples:** Smile in the mirror—do dimples appear?



What Did We Learn Today?

1. Traits are passed from parents to children via genetics. | 2. Dominant traits show up more frequently. | 3. Learned traits come from practice!

Student Practice Worksheet

Grade 4 Science • Lesson: Traits, Inheritance & Genetics

Name: _____

Date: _____

Score: ____ / 15

Question 1: What are **inherited traits**? Define in your own words.

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Question 2: Classify the following as either an **Inherited Trait** or a **Learned Trait**:

(a) Brown eyes: _____ (b) Playing the Tabla: _____

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Question 3: Ananya has dimples just like her grandmother (Dadi). Is having dimples an inherited trait or a learned skill?

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Question 4: Name the scientist who studied trait inheritance using pea plants.

.....

Question 5: Which type of trait is stronger and shows up more often in a family: **Dominant** or **Recessive**?

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Question 6: If both of your parents have dark brown eyes, which eye color are you most likely to inherit?

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Question 7: True or False: Learning how to dance Bharatanatyam from your grandmother is an inherited genetic trait.

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Question 8: Look in a mirror or ask a classmate: What are two physical traits you can observe about yourself?

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Question 9: Between **Curly Hair** and **Straight Hair**, which one is generally considered a dominant trait?

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Question 10: Why can two siblings from the same family look different from each other?

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Question 11: Is riding a bicycle an inherited trait passed down from parents or a learned skill?

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Question 12: Explain the difference between **free earlobes** and **attached earlobes**. Which one is dominant?

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Question 13: Can tongue rolling be improved by genetic inheritance or is it purely a skill taught in school?

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Question 14: Give two examples of **learned skills** that you have picked up from your family members or teachers.

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Question 15: Home Mission Challenge: Identify one trait you share with your mother/father and one trait you share with your grandfather/grandmother.
